PP-S-002128(Army-GL)
June 30, 1982

INTERIM FEDERAL SPECIFICATION

SHRIMP NEWBURG, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by the US Army Natick Research and Development Laboratories, Natick, MA 01760, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers shrimp newburg thermostabilized tray pack cans.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

EE-S-631	-	Spices, Ground and Whole, and Spice Blends
TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

FSC 8905

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Military Specification:

MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence Items; General Specification for

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Canned Foods for the Armed Forces

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Department of Health and Human Services

Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 100-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture

U.S. Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Food Quality Assurance Division, Agricultural Marketing Service, U.S. Department of Agriculture, South Building, 14th and Independence Avenue, SW, Washington, DC 20250.)

U.S. Standards for Grades of Cheddar Cheese

(Application for copies should be addressed to the Dairy Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Department of Commerce

U.S. Standards for Grades of Frozen Raw Headless Shrimp

(Application for copies should be addressed to the U.S. Department of Commerce National Oceanic and Atmospheric Administration, National Marine Fisheries Service, National Seafood Quality and Inspection Laboratory, P.O. Drawer 1207, Pascgoula, MS 39567.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Cheese Institute

Cheese Color Standards

(Application for copies should be addressed to the National Cheese Institute, 110 North Franklin Street, Chicago, IL 60606.)

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Suite 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4, 6.1 and 6.4).

3.2 General. The tray pack product shall be shrimp and rice in cheese sauce.

3.3 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.3.1 Shrimp. Shrimp shall comply to the U.S. Grade B or better of the US Standard for Grades of Frozen, Raw, Headless, Shrimp. The species shall be Rock Shrimp (Sicyonia Brevirostris) and shall be 36 to 42 count per pound. The shrimp shall be peeled and deveined in accordance with industry practice.

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3.3.2 Cheddar cheese. The cheese shall have a well-defined Cheddar flavor and shall be U.S. Grade B or better. All undersirable surface defects shall be removed prior to further processing. The color shall be within the range of No. 5 to No. 8 when compared to the National Cheese Institute's (NCI's) Cheese Color Standard.

3.3.3 Dehydrated Cheddar cheese. Dehydrated cheese shall be derived from U.S. Grade B or better, extra sharp, aged Cheddar cheese. The dehydrated cheese shall be uniform in composition and appearance, free from lumps that do not fall apart under light finger pressure, free from burnt specks and foreign material. The hydrated product shall be free from rancid, stale, oxidized, tallowy, bitter, fishy, soapy, scorched or other objectionable flavors and odors. The flavor shall be that of a typical sharp-cured processed Cheddar cheese. The dehydrated cheese shall not contain more than 3.0 percent moisture and 5.0 percent salt. The milkfat shall not be less than 48.0 percent. One-half of the dehydrated cheese shall be derived from uncolored cheddar cheese and the other half shall possess a deep orange color.

3.3.4 Cornstarch. Cornstarch shall be white, odorless, finely pulverized, modified waxy maize food starch for use in canned foods. The starch shall be bland with essentially no cereal or starchy taste, and shall comply with the bacteriological standard of MIL-STD-900.

3.3.5 Whitener. The whitener or powdered vegetable shortening shall be a white to light cream colored powder possessing a bland, clean, dairy flavor and the moisture content shall not exceed 2.0 percent.

3.3.6 Salt. Salt shall be non-iodized, white, refined, sodium chloride with or without anti-caking agents.

3.3.7 Sugar. Sugar shall be refined, sucrose, liquid or granulated, cane or beet sugar or both. The sugar shall comply with the bacteriological standard of MIL-STD-900.

3.3.8 Flour. Flour shall be white (bleached) general purpose, wheat flour, free from mold and free from lumps that do not break under light finger pressure. In addition, flour shall comply with the bacteriological standard of MIL-STD-900.

3.3.9 Sherry flavor. Sherry flavor shall be a natural wine flavor with alcohol and excess water removed.

3.3.10 Mustard flour. Mustard flour shall be a bright yellow powder prepared from a blend of powders derived from milling the endosperm of Brassica hirta Moench and Brassica juncea with or without the removal of a portion of the fixed oil. Volatile oil content shall be not less than 0.4 percent.

3.3.11 White pepper. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or outer and inner coverings have been removed. Volatile oil content shall be not less than 1 percent.

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3.3.12 Citric acid. Citric acid shall be colorless translucent crystals or a white, granular to fine crystalline powder. It shall be odorless and possess a strong acid taste.

3.3.13 Rice. Rice shall be long grain enriched parboiled, rice. Moisture content shall not exceed 13.0 percent. Broken kernels shall not exceed 4.0 percent.

3.4 Sauce preparation. The sauce shall be formulated and prepared as follows:

Ingredients	Percent
Water	75.59
Cheddar cheese	9.85
Cheddar cheese, dehydrated, colorless	3.37
Cheddar cheese, dehydrated, colored	3.38
Cornstarch	2.00
Whitener	2.00
Flour	1.45
Salt	1.10
Sugar	0.55
Sherry flavor	0.50
Mustard flour	0.15
White pepper	0.04
Citric acid	0.02
	100.00 percent

a. Make a starch slurry with the cornstarch, flour, and part of the water. The starch slurry shall be free of lumps.

b. Add the remaining water to a steam-jacketed kettle. Warm slowly as the dehydrated cheese, whitener, salt, sugar, and citric acid are added. Heat this mixture to 140°F. Add the Cheddar cheese and stir until the cheese is melted.

c. Add the starch slurry, sherry flavor, mustard flour, and white pepper, stirring constantly to prevent lumping.

d. Heat the sauce to 180°F and hold for approximately 5 minutes at this temperature. The sauce shall be smooth in texture and have thickened. Cool to 150°F.

e. The volume of the final mixture shall be adjusted to compensate for evaporation loss during heating.

f. The sauce shall be used within 4 hours after cooking and held at a temperature of not less than 150°F during this interval.

3.5 Shrimp preparation. Shrimp shall be raw, headless, peeled and deveined, whole shrimp. The shrimp shall be filled into the tray pack can without any further processing. The temperature of the shrimp shall not exceed 50°F prior to heat processing into the finished product.

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3.6 Rice preparation. The parboiled rice shall be blanched in hot water and cooled by rinsing with cold water prior to filling into the tray pack cans. The blanched and rinsed rice shall have a moisture content of 52 to 54 percent.

3.7 Tray pack filling and sealing. Each tray pack can shall be filled with approximately 29 ounces of shrimp, 9 ounces of blanched rice, and 68 ounces of sauce such as to conform to the following requirements:

a. The temperature of the sauce at time of filling shall be not less than 150°F.
b. The temperature of the rice at the time of filling shall be 45°F or below.
c. The average net weight per can shall be 106 ounces minimum.
d. No individual can shall contain less than 104 ounces of product.
e. No individual can shall contain less than 16 ounces of drained weight of shrimp.
f. No individual can shall contain less than 22 ounces drained weight of rice.
g. The can shall be hermetically sealed with a minimum of 10 inches of vacuum immediately after filling.

3.8 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting at 240°F in water until a sterilization value (F_o) of not less than 6.0 nor more than 8.0 has been achieved at the coldest heating point of the retort.

3.9 Salt content. The salt content of the finished product shall be not less than 1.0 percent nor more than 1.5 percent.

3.10 Finished product requirements. Finished product requirements shall comply with the following:

a. There shall be no foreign material such as, but not restricted to glass, paint, metal, dirt, insects, insect parts, wood or paper.
b. There shall be no foreign odor or flavor such as, but not restricted to burnt, scorched, stale, sour, rancid, musty or moldy.
c. There shall be no foreign color such as, but not restricted to black, purple or green.
d. Shrimp shall not be soft, mushy, or rubbery.
e. Rice shall not be hard or mushy.

3.11 Plant qualification. The product shall be prepared, processed, and packaged under sanitary conditions in accordance with good manufacturing practice. The product shall be processed in establishments meeting the sanitary requirements of the applicable government inspection agency.

3.12 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this specification. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

4.3 Plant qualification. The product shall be unacceptable if not produced in plants which currently meet the requirements of 3.10.

4.4 First article inspection. When a first article is required (see 6.1), it shall be examined for the defects specified in table I and tested for the characteristic specified in 4.8.3. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.5 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.6 Ingredient examination. Conformance of ingredients to identity and condition requirements shall be determined by examination of pertinent labels, markings, U.S. Grade certificates, invoices, certificates of analysis, or other valid documents. In addition each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity or condition requirement shall be cause for rejection of the ingredient lot or of any involved product.

4.7 In-process examination. In-process examination shall be performed to determine conformance to the ingredient preparation, product processing, can interior coating, and sealing and retorting requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature and formulation; certi-ficates of conformance; or other valid documents shall be cause for rejection of the involved product.

4.8 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

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4.8.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 0.1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

4.8.2 Product inspection. Eight tray packs shall be selected at random from the lot. The eight tray packs shall be heated at least 15 minutes in 140 to 150°F water, opened, and inspected for the defects listed in table I.

a. If no defects are found, the lot shall be acceptable.
b. If two or more defects are found, the lot shall be rejected.
c. If only one defect is found, a second sample of eight tray packs shall be selected at random from the lot, heated as above, and inspected for the defects listed in table I. If any defect is found in the second sample, the lot shall be rejected.

TABLE I. Product defects inspection 1/

Category	Defect
Major	
151	Washed and drained weight of shrimp is less than 16 ounces. 2/
152	Washed and drained weight of rice is less than 22 ounces. 2/
153	Presence of foreign color, such as, but not limited to black, purple, blue, green, or white.
154	Shrimps soft, mushy, or rubbery in texture.
155	Rice hard or mushy in texture.

1/ The presence of any foreign material (e.g. glass, dirt, insect parts, hair, wood, metal, etc.) or foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale, etc.) shall be cause for rejection of the lot.

2/ To determine drained weight in a can, the free liquid shall be poured off and the remaining contents shall be poured into a flat bottom container. A minimum of three times the tray pack can's volume of 190 to 212°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated such as to liquify rendered fat and to remove the sauce without breaking the shrimp or rice. The contents shall then be poured into a U.S. Standard 1/4 inch sieve in a manner that will distribute the product over the

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sieve without breaking the shrimp or rice. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted slightly and allowed to drain at least 2 minutes before determining the drained weights of shrimp and rice by subtracting the sieve tare weight from the gross weight. The drained weights shall be reported to the nearest 0.1 ounce.

4.8.3 Salt content testing. Nine tray packs shall be selected from the lot and distributed as follows:

- Three for laboratory analysis.
- Three for submission to the contractor.
- Three for retention by the inspector as a reserve sample.

The three tray packs for laboratory analysis shall be tested for salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, except that preparation of the sample shall be as follows. The three unopened tray packs shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and a composite of the entire contents shall be blended in a Waring blender or equivalent to form the sample. The lot shall be rejected if the salt content in any individual can is less than 1.0 percent or more than 1.5 percent. Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of non-compliance with the salt content requirement. When the reserve sample is analyzed, the analyses for salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.9 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers.

4.10 Shipping container examination. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in construction, closure, and reinforcement shall be in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible. Minor: (1) Other required markings missing, incorrect, or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) When applicable, reinforcement with nonmetallic strapping or tape is not used. Level C shipping containers shall be examined only for the marking defects heretofore specified.

4.11 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

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5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 106 ounces of the food product, sealed, and thermoprocessed as specified in 3.6 and 3.7. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient weight and temper to protect the product during shipment and storage. The entire inside area of the can shall be coated. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed, closed and reinforced in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pads shall be full length and width less approximately 1/4 inch in each dimension. The box liner and pads shall be fabricated of grade V3c fiberboard.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.2.1 Nonmetallic reinforcement of shipping containers. When specified (see 6.1 and 6.3), the shipping container specified in 5.2.2 shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The shipping container shall be in accordance with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. For distribution to Navy, when unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each package shall be labeled to comply with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder. Paper labels are not permitted. The name of the product or abbreviation thereof and date of pack (day, month, year) shall be embossed or lithographed on the end (lid) of the can.

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TO HEAT

a. To heat in water: Submerge unopened can in boiling water. Simmer gently 35 minutes. Remove from water and open lid for serving.

b. To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 35 to 40 minutes until contents reach an internal temperature of 165°F.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1, 4.4 and 6.4).
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) When nonmetallic reinforcement is required for level B packing (see 5.2.2.1).
(e) Type and class of unit load when unit loading is required (see 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 Selection of nonmetallic reinforcement. The nonmetallic reinforcement specified in 5.2.2.1 should be selected for shipping containers that are destined for transfer at sea or other specific overseas operations.

6.4 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of three tray cans. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

Preparing activity:

Army - GL

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